B.Sc. (Hons) in Software Development

CraveControl: A Web Application for Nicotine Cessation

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Abstract

Vaping was introduced to consumers with the initial ideology of being replacements to cigarettes and to aid those who aim to halt their intake of nicotine products. However, in recent years, vaping in its own right has developed its own culture and its popularity significantly increased, especially among those in the younger generations. With this trend increasing in popularity, it is losing its original concept of being a quitting aid. This has led to people who have not previously encountered vapes and e-cigarettes to be exposed to them and possibly leading them to try these products out. Since there is an increase of intake, there is a direct correlation of increase of those who wish to quit their intake of nicotine products. This dissertation will be discussing CraveControl, a web application that was recently developed. It will also investigate different cessation methods, with a focus on financial accumulative savings and data visualisation of self-tracked cravings levels.

CraveControl was designed and implemented as a web application that allows users to log cravings, track them over time and visualise their craving patterns. It tracks the accumulative savings from avoided nicotine product purchases and is displayed to the user. The system was evaluated through small user-based usability tests to demonstrate its effectiveness and that the overall functionality works as expected. These tests included the effectiveness of aiding those who want to quit through self-awareness and financial motivation. Results have indicated that users found that the visualisation techniques used were helpful in reinforcing behavioural awareness and app engagement within their cessation journey.

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Chapter 1

Introduction

VAPES, e-cigarettes and nicotine pouches were originally introduced as a secondary option for traditional smoking, being marketed as a less harmful option to cigarette and tobacco products to help smokers in reducing their nicotine intake or completely quitting. However, with their rapid popularisation, vaping has become a less stigmatised alternative to smoking rather than an aid in diminishing smoking, where the adverse health effects have negligible influence on the decision to begin or continue to vape. Alongside the newly available nicotine pouches, this has created an intrigue for young people as they are more susceptible to marketing and cultural shifts, which creates a great health concern for their generations.

In 2019, a survey conducted on a group of 15 to 16 year old students showed that 1 in 2.6 students had used a vape or e-cigarette at one time [3], with 2 in 3 students trying it "out of curiosity" as their main reason. Lately, in 2025, it was documented that in a government survey conducted on 15 to 24 year olds, 1 in 5.6 people previously vaped or currently vape [4]. As it continues to become increasingly normalised in young generations, it shows the importance of developing new technological recovery and support tools in aid of cessation that focuses specifically on vaping and nicotine pouch behaviour and demographic.

1.1 Nicotine Consumption Behaviours and Patterns

Every nicotine user has a unique consumption pattern, which is shaped by routines, individual circumstances and environmental factors. Since vapes and e-cigarettes do not require a lighter or ignition and are compact in design, vaping is more convenient than smoking a traditional cigarette. The situation for nicotine pouches is similar as they are stored in a compact box with a separate compartment to store used pouches, removing the need to be near a place of disposal. This convenience can lead to an increase of passive intake, alongside the physical ease of using the device, where users may not necessarily realise how frequently they consume nicotine on a daily or weekly basis. The absence of needing multiple devices for nicotine intake that is required for traditional cigarettes makes the activity less conscious which makes it difficult to track without a tracking system.

Furthermore, ease of access to nicotine products intensifies this issue. Vapes and nicotine pouches being accessible in newsagents and corner stores contributes to the increase of popularity, which makes them the second most popular retail outlet to sell vapes [4]. Combined with their relatively low cost, this can reduce the impact of purchasing nicotine products to an insignificant purchase, contributing to the lack of self-awareness of the quantity of products they consume. These factors make the habit of nicotine consumption challenging to quit without support.

1.2 Obstacles to Quitting

Quitting nicotine presents many obstacles to users who seek to stop their consumption, which can appear in multiple forms. These can be financial, behavioural and social aspects. The most significant of the obstacles to overcome is nicotine addiction. As a result of continuous and regular intake, the user can develop a physical dependence and addiction to the nicotine contained in vaping pods, disposable vape devices [5] and nicotine pouches. This addiction is further fuelled by negative reinforcement [6] as craving levels are at their highest right before smoking. This creates a dependence as craving levels immediately drop after nicotine intake and reduces the negative mood caused by those cravings, which is comparable to nicotine pouches and vapes due to nicotine content.

Outside of physical addiction, nicotine products can be developed as a habit and become integrated into daily routines and social situations. Through this research, there is a lack of vape or nicotine pouch focused cessation applications and tools, contributing to the challenge of quitting. While there is an abundance of smoking focused cessation applications and tools, there are differences to vaping and pouches that some of those tools may not be as effective to those who aim to quit. These include intake behaviour, frequency and demographics, with the younger generation being more likely to vape than older generations. Due to this, there is a need for the development of a dedicated solution that focuses on modern nicotine products and to the corresponding age groups.

1.3 Project Aim and Motivation

This dissertation delves into CraveControl, a web application developed for the purpose of supporting nicotine product cessation. It achieves this using two main components: the visualisation of self-tracked craving levels over a period of time, and monitoring of accumulative savings gathered by avoiding product purchases. The motivation for this research is the fact that there is a lack of general nicotine cessation tools overall, and ones that also implement both financial monitoring and craving visualisation, relative to smoking-focused tools.

The core theory of this paper is that there are a multitude of different methods to enable consumers of nicotine to quit using self-awareness and financial accountability. These include visual representation of their craving patterns and a graphical representation of savings accrued through preventing product purchases. Using the actual product price, instead of an estimate, provides more personalised and accurate feedback to contextualise financial gains.

This research paper is further motivated by personal experience and encounters that depicted the difficulties that people faced when attempting to quit vaping or pouches, which emphasised the lack of resources and platforms for this expanding issue.

1.4 Research Questions

This paper investigates the following research questions:

RQ1: What technical implementations of data visualisation are best matched for ease of usability?

RQ2: Which data visualisation technique is best suited to accurately display self tracked craving data?

RQ3: What is the most effective way to implement personal financial monitoring and display accumulative savings?

1.5 Paper Focus

The primary focus of this paper is the design, implementation and evaluation of CraveControl. CraveControl is a vape cessation web application that implements the visualisation of self-tracked cravings and accumulative financial monitoring based on product selection. From research performed, there is a scarcity of cessation applications that apply these two features in a single tool. Moreover, the application targets nicotine cessation rather than traditional cigarette cessation, which addresses a gap in current studies and literature for technological support tools for this growing issue.

Another focus of this paper is the comparison of multiple existing cessation applications and different data visualisation techniques, which provides insight into the design choices made for the development of features and tracking visualisation in CraveControl. The aim is to investigate whether these two features combined are effective for nicotine consumers in aid of quitting.

1.6 Project Repository

The project GitHub repository can be found here:
`https://github.com/natashazywczyk/FinalYearProject`

The repository consists of:

- A `README.md` file that describes the premise of the project and how to deploy the application locally.
- Within the `README.md` file there is a link to a screencast of the application: Screencast linked here.
- Folder `CraveControl` containing all files necessary for the application

When making a Quasar project, the default setup is to make a folder within a repository. Within the `CraveControl` folder, the main components are:

- `scraper`: Contains all web scrapers used for images, names and prices of nicotine products.
- `src`: Contains frontend visualisation, components and handlers between frontend and back-end.
- `supabase`: Contains previous and current migrations of the Supabase database.

1.7 Paper Contents

Below is the breakdown of the five remaining chapters included in this dissertation:

- Chapter 2: Describes the methodology used to accumulate resources and applications, using exclusion and inclusion criteria.
- Chapter 3: Analyses related works and research to the topic of this dissertation and the features they provide or lack.
- Chapter 4: Displays and discusses the system design and architecture of the web application, including frontend and back-end.
- Chapter 5: Evaluates the project and reports on the strengths and weaknesses of the application.
- Chapter 6: Compiles the paper's findings, obstacles and possible future work.

Chapter 2

Methodology

2.1 Overview

This chapter outlines the research methodology used to gather resources and the deciding factors deployed for selecting relevant academic material for this paper. Below are the databases used to identify eligible studies, that adhere to the inclusion and exclusion criteria described, and search strings used to find related papers. There is a summary provided at the end of this chapter showing the overall search results and identifies the challenges that arose while looking for supporting materials.

Since this paper covers multiple disciplines like physiology, psychology and technology, the criteria was strictly adhered by to prevent diversion from the subject of the paper. Data visualisation, frameworks and personal financial monitoring methods are found in the scientific technological databases, while health risks, cessation approaches and mobile health applications are found in medical databases. The methodology was implemented for all sources cited in this paper.

2.2 Database Selection

Three main databases were used, one medical and two technical. IEEE Xplore and Google Scholar were used for technical background and implementation. To gather statistical and correlating information on vaping and its effects, PubMed was used. PubMed provided papers that discussed vape statistics among younger generations and its growing prevalence. Each database has its own role in identifying corroborating materials.

2.2.1 IEEE Xplore

IEEE Xplore was the primary database used for finding academic, peer-reviewed technical studies regarding different data visualisation techniques, personal financial systems and mobile cessation applications and the techniques they utilised. Since IEEE Xplore is a digital library consisting of papers in computer science, electrical engineering and electronics, it provides access to journals, articles and conference papers. Papers indexed by IEEE Xplore are added once they are accepted and published through IEEE-sponsored means. These papers included different technical implementations that could be used in the development of CraveControl.

The advanced search feature was consistently used for the research of technical techniques for CraveControl and information for this paper. With its wildcard operations and Boolean operators, it provided functionality to refine the search for papers with a specific focus.

2.2.2 Google Scholar

Google Scholar was used as a secondary database to increase the scope of the search results for technical papers stored by IEEE Xplore. Unlike IEEE Xplore, there is a greater amount of papers returned and they may not be as targeted since Google Scholar has a variety of material types including but not limited to books, theses and grey literature. Papers indexed by Google Scholar are sourced from academic institutions and online repositories. The advanced search feature was used similarly to IEEE Xplore's search. Google Scholar was primarily used to identify sources that may not have been indexed by IEEE Xplore.

Google Scholar was also used to find supporting survey papers on vaping statistics and its omnipresence in younger generations.

2.2.3 PubMed

PubMed was specifically used to find medical papers that provided insight for background context and statistical data highlighting the popularity of vaping among young people. Studies that focused on possible health risks and physiological effects of vaping were referenced for better context to the growing public health concern discussed at the beginning of this paper. PubMed also provided multiple papers about mobile cessation applications, however, it was not used for searching for technical implementation papers.

2.3 Search Strategy

The primary databases utilised for sourcing technical papers were IEEE Xplore and Google Scholar, while PubMed was used to discover contextual medical academic information. The medical papers gathered contained information pertaining to vaping physiological effects and mobile health applications. Searches on Google Scholar used the default article filter, as well as its advanced search. IEEE Xplore was used consistently with continuous use of its advanced search functionality, using Boolean search strings and wildcards that heavily reduced the results to papers that were relevant to this paper.

The decision was made to utilise multiple, shorter search strings that were more targeted than a longer singular query. At the beginning when trying to gather references, long queries were found to disregard and exclude papers that were relevant due to a difference in terminology. One of the main differences was the inconsistency in spelling between British and American spellings. In papers regarding data visualisation, both 'visualisation' and 'visualization' were used interchangeably which affected the search results. This highlighted the need for the wildcard operator to allow both spelling variations in the one search query result.

An example search string applied for Google Scholar is as follows:

data AND vape AND (visualization OR visualisation) AND tech AND financial AND personal*

An example of a search string applied for IEEE Xplore is as follows:

data AND (visualis OR visualiz*) AND financial AND (vape OR smok*) AND techni**

The wildcard operator applied in IEEE Xplore search strings accommodates for the interchangeability of related terms like 'visualisation', 'visualization' and 'visualised'. The Boolean OR operator was used for terms similar to each other. The string 'vape OR smok*' was utilised as there is a notable gap between the amount of smoking cessation applications to vape focused applications.

2.4 Inclusion and Exclusion Criteria

To ensure that the sources collected were of expected relevance and quality, there was a set of inclusion and exclusion criteria outlined and followed by. As part of the inclusion criteria, there was an importance to only include journals, articles and conferences that were peer-reviewed and from within the last six years, spanning from 2020 to 2026. Articles had to be in English to be considered as a source for the research, as part of the inclusion criteria. Grey literature, which includes blogs, non-peer-reviewed papers and websites, were specified as part of the exclusion criteria.

The reasoning behind the choice of only using papers from the last six years was to only include papers that cover the current state of frameworks and technologies of technical papers and to get the most recent medical statistics. Since there are mobile health applications being released per year, as well as continuous development on visualisation technical implementation and frameworks, referencing papers predating 2020 increases the risk of the material no longer being applicable to current practice.

Table 2.1 presents a summary of the inclusion and exclusion criteria applied to all three databases.

Criteria	**Inclusion**	**Exclusion**
Publication type	Peer-reviewed journals, articles, conference papers	Blogs, opinion-based papers, unverified websites, non-peer-reviewed articles
Publication date	2020 to 2026	Prior to 2020
Language	English	Non-English
Topic focus	Technical implementation, mobile health applications, data visualisation, financial monitoring	Papers on application reviews with lack of technology discussion
Relevance	Relevant to vaping/smoking cessation, data visualisation, personal financial tracking	Discussing unrelated medical conditions, industry scale financial systems

Table 2.1: Inclusion and exclusion criteria

2.5 Search Results

Due to using multiple search strings to find corroborating materials, the number of papers returned fluctuated. This shows the difference of how academic items are indexed across multiple databases. The results of the example search string above for Google Scholar resulted in approximately 15,800 articles, while the example search string for IEEE Xplore resulted in 2,439 articles. This demonstrates the broadness associated with Google Scholar's indexing compared to that of IEEE Xplore. This further shows the reasoning for using IEEE Xplore as the primary database used for collecting technical resources for the purpose of this paper.

The comparison in the amount of articles returned by both Google Scholar and IEEE Xplore demonstrates their respective scopes and search accuracy. With IEEE Xplore having a smaller article result set, it was decided that it would be the main source for technical articles as it was more concise.

Table 2.2 presents a summary of the searching process applied across all three databases.

Database	**Purpose**	**Example results returned**	**Papers retained**
IEEE Xplore	Technical literature: visualisation, mobile health, cessation apps	2,439	Majority of retained technical papers
Google Scholar	Supplementary broader literature	∼15,800	Small number of sources not found in IEEE Xplore
PubMed	Contextual medical and health statistics	Differs per string	Contextual background sources

Table 2.2: Search results between the different databases

2.6 Summary

This chapter displays the approach behind choosing databases, gathering relevant resources and the filtering methods applied in finding articles. The three overall databases used were: IEEE Xplore, Google Scholar and PubMed. In the technical databases, multiple shorter, concise search strings involving Boolean operators and wildcards were used in filtering articles to those that fit the research topic of this paper. Only papers that followed the inclusion criteria were included, while those that fell under the exclusion criteria were discarded. This criteria was followed closely to ensure quality and consistency throughout this dissertation.

2.7 Software Development Methodology

CraveControl was developed using an iterative, agile approach, building the application incrementally with multiple cycles which were shaped and adapted based on technical user feedback. Instead of following a rigid sequential plan, development progressed in informal sprints where a set of features were implemented, tested and finalised before moving onto the next. This allowed the scope and design of the applicaiton to evolve naturally as new reuqirements emerged.

Requirements for the project were gathered from the technology review in Chapter 3, which identifies gaps in current technological tools and helped identify the main features, and multiple rounds of peer testing. The peer testing uncovered additional features that were not initially identified during development. The most significant of these was the sidebar menu panel, which was suggested during the first round of peer testing and included in the final version of the application was released. This demonstrates how peer feedback was directly mapped to specific development iterations rather than being collected and addressed at the end.

The application source code is managed using Git version control, hosted in a GitHub repository [7]. GitHub Issues was used throughout development to track bugs, feature requests, and outstanding tasks. GitHub Project was also utilised to keep track of issues and feature integration. A Kanban project board was selected as the most appropriate format for tracking development progress. It provided a clear visual overview of tasks in backlog, in progress and completed states.

Testing consisted of incremental developer functional testing throughout development and multiple rounds of peer-based user testing on both incomplete and completed versions of the application. Further details are discussed in Chapter 5. The usability evaluation framework applied in Chapter 5 follows Nielsen's usability heuristics [8], which were identified through independent research into established usability evaluation methods rather than the primary database search strategy described above.

Chapter 3

Technology Review

This chapter reviews the technologies and existing applications that are relevant to the development of CraveControl. There are three areas covered as part of the investigation: data visualisation techniques and technologies, existing mobile health applications, particularly cessation focused, and personal financial monitoring systems utilised. The reasoning of the review is to identify current applications of this type and observe the features used and their limitations. The review will provide justification for design choices made in the development of CraveControl. There is a discussion of the overall findings of the reviewed applications at the end of the chapter.

3.1 Data Visualisation

Data visualisation is a well-established field covering numerous techniques and technologies for presenting data in a readable and appealing form. Its primary use in health applications is to transform logged data and present it into visualised patterns of behaviour which can aid in decision-making. Due to there being a large availability of many different variations of data visualisation tools, there are technologies that are more interactive and flexible. There are dedicated libraries in languages like JavaScript, Python and R [9]. There are platforms dedicated to visualisation without prior programming knowledge.

A major challenge within this field is with scalability. As datasets alter over time, presenting updates accurately becomes more difficult to maintain [9]. Progressive visualisation has emerged to specifically address this limitation, which improves the visualisation of data as data logged fluctuates [10]. This is important for cessation applications that rely on data visualisation of health statistics and financial data to remain readable as user data increases.

3.1.1 JavaScript Visualisation Frameworks

For web-based and mobile compatible applications, JavaScript libraries are the most prominent choice for implementing interactive data visualisation. Chart.js [11], D3.js [12] and ApexCharts [13] are the libraries that are the most widely used. Each of these libraries offer a different degree of flexibility, simplicity and interactivity.

Chart.js is an open-source library that provides an API for building multiple chart types including bar, line and pie charts using the HTML5 Canvas element [14]. Due to this, it is well suited to applications that implement standard chart types that do not need additional customisation. D3.js is a highly flexible low-level library that applies data to Document Object Model elements, allowing for more complex data visualisation. However, there is a learning curve and large library, though providing more options, makes it less desirable for rapid development of standard chart types in mobile health applications.

ApexCharts is another open source charting library that provides interactive SVG-based charts. These charts support interactive tools like zooming and tooltips. It provides native component integrations for many JavaScript frameworks including Vue.js [15]. This ability makes it suitable for implementation within applications using Quasar Framework as it is component-based and built on Vue.js. The interactive capabilities of the charts, like the hover data inspection, provide more functionality to the user to view their logged data with further detail.

3.2 Mobile App Visualisation

Mobile devices have an inherent restriction to data visualisation due to the limited screen size and with touch gesture interactions, which require visualisation libraries with mobile compatibility to be implemented. Research conducted on mobile health data visualisation discovered that bar and line charts are most widely implemented and sought after in those applications [16, 9] due to these charts usually being supported across many mobile visualisation libraries. It was found that users prefer simple, readable charts over complex interactive visualisations [16]. This contributed to the decision to implement a clear line chart to display user data in CraveControl using ApexCharts [13], which provides mobile-responsive rendering and touch-compatible interaction.

3.3 Wearable Sensor-focused Applications

Outside of software approaches, there is a separate branch of health monitoring research that has experimented with wearable sensors being used to collect passive physiological data from users. While this approach is outside of the scope of CraveControl, which is specifically designed for accessibility as a web application requiring no additional hardware, it is worth acknowledging as a sector in the field of health tracking tools. It aids in contextualising the design choice to aim for software accessibility over sensor-based data recording.

A prime example of this category is *HOT Watch* [17] which is an IoT wearable monitoring system found through IEEE Xplore. The application is used to track health statistics of the person wearing the sensor including oxygen rate, heart rate and body temperature. By combining the calculations of these readings, the system displays to the user whether their health metrics are abnormal or normal [17]. It has been demonstrated that vaping and smoking each affect a person's heart rate [18, 19] and oxygen level [20] due to the presence of nicotine. This indicates that sensor systems, such as *HOT Watch*, could be used to watch physiological effects that can present themselves with nicotine intake.

However, *HOT Watch* is strictly a read-only health monitoring system and lacks cessation-based functionality. It does not allow for user input, which in turn does not allow for craving tracking and financial monitoring. The dependency on a dedicated wearable device reduces the accessibility to that of a web application. For CraveControl, it was decided to not implement sensor compatibility and integration, focusing on a self-tracking application that requires only one device, either a mobile device or computer. This allows for a wide user base without requiring extra devices to use all the applications functionality.

3.4 Mobile Health Applications Use Cases

3.4.1 Sense2Quit

Sense2Quit [1, 21] is a native mobile health application discovered through research that implements both financial monitoring and data visualisation, that includes wearable sensor integration. The application is cigarette cessation focused, however the approach to financial monitoring and data visualisation is viable to the nicotine-centric context which makes it relevant to this paper.

The application visualises both cigarette consumption and accumulative financial cost of each individual cigarette smoked. As shown in Figure 3.1, the price of each cigarette is calculated from the price of the box, from the manual input from the user. The data on cigarette consumption is displayed on a bar chart, with the x-axis representing each day and the y-axis representing the amount of cigarettes smoked (see Figure 3.2). A separate tab on the bar chart visualises the money spent of cigarette products, depicting the correlation between consumption and financial spending. The user's financial data is visualised graphically and numerically, with the figures also displayed in a table above the chart, as shown in Figure 3.3.

During the development of *Sense2Quit*, there were changes iteratively made to the data visualisation [1], which involved removing unnecessary decimal values from the axis labels which introduced visual noise. This demonstrates the importance of simplicity and clear visuals in health data visualisation. This analysis contributed to the design decisions made in the development of CraveControl.

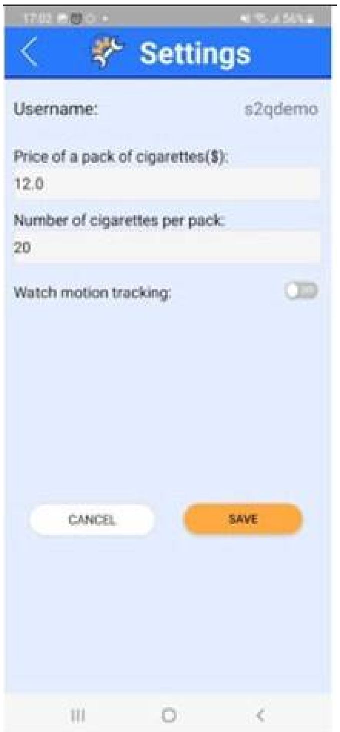

Figure 3.1: Price of cigarette box manually entered [1]

Figure 3.2: Bar chart visualising cigarettes smoked over a week [1]

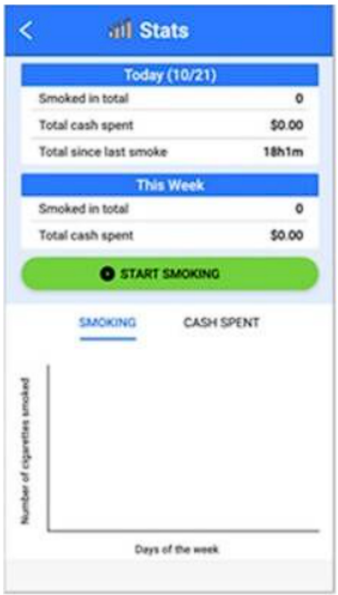

Figure 3.3: Final round changes to app screens organized by usability round [1]

3.4.2 QuitGenius

QuitGenius [2, 22] is another native mobile smoking cessation application that implements data visualisation both numerically and graphically that is different to *Sense2Quit*. While *QuitGenius* is a smoke-centric cessation application [22], the method of data visualisation it utilises is relevant to this research as a point of comparison.

It prioritises positive reinforcement through cognitive behavioural therapy (CBT) based approach [23] (see Figure 3.5). Instead of presenting the data graphically, it displays logged results across multiple categories using both quantitative and qualitative data, as shown in Figure 3.4. This numerical and statement-based way of visualisation is different to the bar chart approach of *Sense2Quit*. It shares a similarity with *Sense2Quit* where it gathers data from the user on a daily basis in order to help identify behaviour patterns through qualitative data [2].

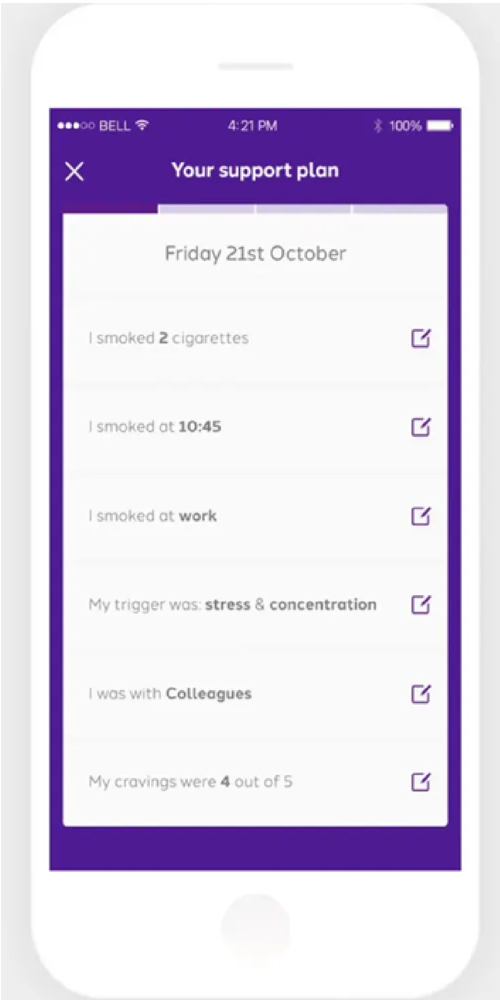

Figure 3.4: Capture taken from *Quit-Genius* website [2]

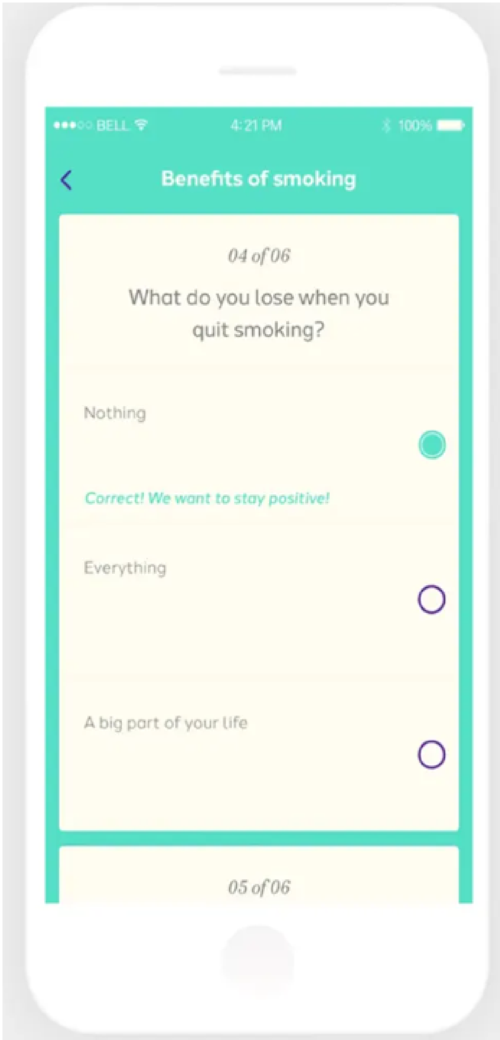

Figure 3.5: Capture taken from *Quit-Genius* website [2]

QuitGenius, although it provides a general estimate of financial benefits accumulated since the user's quit date [22], it is limited to a generic calculation instead of one based on the user's specific product and its corresponding price. Another obstacle is the fact that it operates on a business-to-business (B2B) model. This means that the application can only be accessed through organisations, like employers or healthcare providers, that hold a license [22]. This prevents individual consumers from using the application, which limits its accessibility to users who are looking for cessation support without going through an organisation.

3.5 Financial Monitoring

3.5.1 Sense2Quit

While doing research for this paper, there was a noticeable lack of academic papers on cessation application that implement accumulative financial monitoring. However, *Sense2Quit* [1, 21] addresses this through a combination of numerical and graphical financial data visualisation. As shown in Figures 3.2 and 3.3, the 'Stats' page displays tables at the top of the screen that includes the daily and weekly accumulative totals of money spent on cigarette products. This shows a pattern and correlation between individual cigarette cost and consumption.

As well as the table displaying the total consumption and cost of the cigarettes smoked, there is a bar chart provided for the amount spent over the duration of a week. As shown in Figures 3.2 and 3.3, there is a tab 'Cash Spent' which displays the information in the tables in graphical format. This presents a spending and consumption pattern to the user, providing personal feedback. However, the financial data in *Sense2Quit* depends on the user manually inputting the price of a pack of cigarettes, rather than automatically incrementing the accumulative total with the price of a pre-selected product. This introduces a dependency on manual input that CraveControl addresses and avoids through product-based pricing.

3.5.2 QuitGenius

QuitGenius [2, 22] provides a general estimate of financial benefits accumulated since the start of the user's quit journey, however it is limited to a generic calculation. This shows a lack of personalised financial feedback and a surface-level implementation of financial monitoring which may not accurately calculate the user's specific nicotine habit.

3.6 Application Comparison

Table 3.1 provides a side-by-side comparison of key features of the two highly comparable cessation applications reviewed in this paper, *Sense2Quit* [1, 21] and *QuitGenius* [2], against CraveControl. *HOT Watch* [17] is excluded from the table as it operates differently to these tools, being a passive wearable health monitor rather than a cessation application.

Feature	**Sense2Quit**	**QuitGenius**	**CraveControl**
General nicotine focus	✗	✗	✓
Platform	Native mobile	Native mobile	Web application
Requires installation	✓	✓	✗
Accessibility	Public	B2B only	Public
Data visualisation	✓(bar chart)	✓(numerical)	✓(line chart)
Financial monitoring	✓(generic)	✓(generic)	✓(personalised)
Self-reported user input	✓	✓	✓
Craving intensity tracking	✗	✗	✓
Product-based savings calculation	✗	✗	✓
Wearable sensor integration	✓	✗	✗

Table 3.1: Comparative analysis of cessation applications and CraveControl

3.7 Technology Summary

The review of existing applications reveals multiple consistent limitations in current cessation tools. The main gap uncovered during research is the complete lack of cessation applications dedicated to general nicotine products, not just traditional smoking. This is despite the growing prevalence of vaping and nicotine pouches, especially among younger demographics as discussed in Chapter 1. Searches returned no applications specifically designed for cessation of these products. The reviewed applications, *Sense2Quit* and *QuitGenius* target cigarette smoking exclusively [1, 22]. The overall literature search showed no equivalent tools for vaping or nicotine pouch cessation. This indicates a major gap in technological support in the current mobile health field, as vaping behaviour patterns, product types and associated demographics contrast substantially from that of traditional smoking.

Furthermore, there is no reviewed application that utilises both craving intensity tracking and personalised accumulative financial monitoring. *Sense2Quit* provides financial data based on cigarette quantity rather than a product selected by the user [1, 21]. *QuitGenius* only provides a generic financial benefit figure rather than a personalised product-based calculation [2, 22]. Of both of those applications, neither of them provides functionality to track self-recorded craving intensity and visualise it over time.

The wearable sensor approach shown by systems like *HOT Watch* [17] demonstrate the capability of capturing physiological data relevant to nicotine use. However, it introduces a hardware entry limitation and dependency. A web application that uses self-tracked data with no additional devices is more widely accessible to users.

Finally, there is an overall lack of technical papers that document the implementation of personal financial monitoring systems in both a cessation context and in general. While researching for this paper, it was discovered that most papers identified relating to financial monitoring focuses on large-scale industrial level. This includes large datasets, fraud detection and business financial system rather than small-scale, personal spending tracking. This gap in literature for personal financial systems is a significant finding, as it suggests that the combinations of personal financial tracking with health behaviour shows an under explored area of research. This lack of comparable implementation means that the financial monitoring feature for savings in CraveControl was created from limited examples in a cessation context, mostly from *Sense2Quit* [1], instead of information of personal financial monitoring systems.

CraveControl addresses these gaps by providing a vape and nicotine product specific cessation tool that implements self-tracked craving level visualisation as a line chart using ApexCharts [13], with personally selected products used to track accumulative financial savings using its corresponding retail price. The design decisions made in the development of these features and functionalities are discussed in Chapter 4.

Chapter 4

System Design

4.1 Overview

This chapter provides an overview of the design and implementation of CraveControl. CraveControl is a web application created and designed to aid users in quitting vaping and nicotine products using visualisation of self-tracked cravings and accumulative financial monitoring.

Below is the technology stack that CraveControl consists of, the system architecture, the authentication system, the database and its table schemas. There is discussion below explaining the technical decisions made in the development of CraveControl and the challenges that appeared as development continued, with the approaches made to overcome them.

4.2 Technology Stack

CraveControl is a full stack web based application that utilises a JavaScript centric stack. Every platform and technology was used to provide users with a responsive and user-friendly cessation application that is data centric. Libraries were selected to ensure multiple device compatibility for ease of access.

Table 4.1 displays a summary of the technologies used for the application.

Technology	Role	Reasoning
Quasar Framework	Frontend UI framework	Built on Vue.js, provides a default and mobile compatible component library and system suitable for a health application used multiple devices
Tailwind CSS	UI Styling	CSS framework used alongside Quasar for custom component styling, no need for creating custom CSS files
Supabase	Backend-as-a-Service (BaaS)	Provides PostgreSQL database, RLS, real-time API
Supabase Auth	Authentication	Manages user sessions, password-based authentication, OAuth Google integration
Google OAuth 2.0	Google login capabilities	Allows users to register and log in using a Google account, increasing usability
GitHub	Version control	Microsoft owned, industry standard version control platform, Kanban project tracking issues
ApexCharts	Data visualisation	JavaScript charting library with Vue.js support, supports interactive line charts, used to display craving levels over time period
Docker	Local development environment	Used to run a local instance of Supabase during development, local testing of database queries without affecting production version
Vercel	Cloud deployment platform (PaaS)	Provides deployment from the GitHub repository, low-latency access, automatic redeployment of app with every commit

Table 4.1: CraveControl technology stack

4.3 System Architecture

CraveControl consists of the client-server architecture, using Quasar Framework [24] which is built on Vue.js [15] as its frontend and Supabase [25] as its PostgreSQL backend over HTTPS. Quasar communicates with Supabase using the JavaScript client library built into Supabase [26]. Supabase acts as an API as well as a database layer which removes the necessity of a custom middleware layer. With its row-level security policies (see Section 4.8.2) in place at the database layer, users can only view their personal information and data available on the application and not that of other users [27].

When the user accesses CraveControl, a new authentication session is created if one is not currently active. If there is no active session, the user is redirected to the landing page, where they are given the option of either logging into an existing account or to create a new one. The default authentication method is registering an email and password, with the option to login using an existing Google account employing the Google Cloud OAuth 2.0 utilities [28]. When authentication is registered as successful, Supabase produces a JSON Web Token (JWT) [29] to the client-side that the JavaScript client library persists and handles. Attached to the JWT is the subsequent database requests which maintains the correct scope of the authenticated user. It improves the flow of user authentication by removing the need for manual token management in the application code. Project management and version control are discussed in Section 2.7.

CraveControl is deployed on Vercel, a cloud based deployment and hosting platform. Vercel is focused on frontend development rather than being a general purpose cloud platform [30]. Whenever a commit is pushed to GitHub [7] to the linked repository, Vercel automatically rebuilds the Quasar web application and shares the resulting output across its global content delivery network (CDN). This ensures that, regardless of geographical location, users will have low-latency access to the application.

The system architecture is depicted below as a diagram in Figure 4.1.

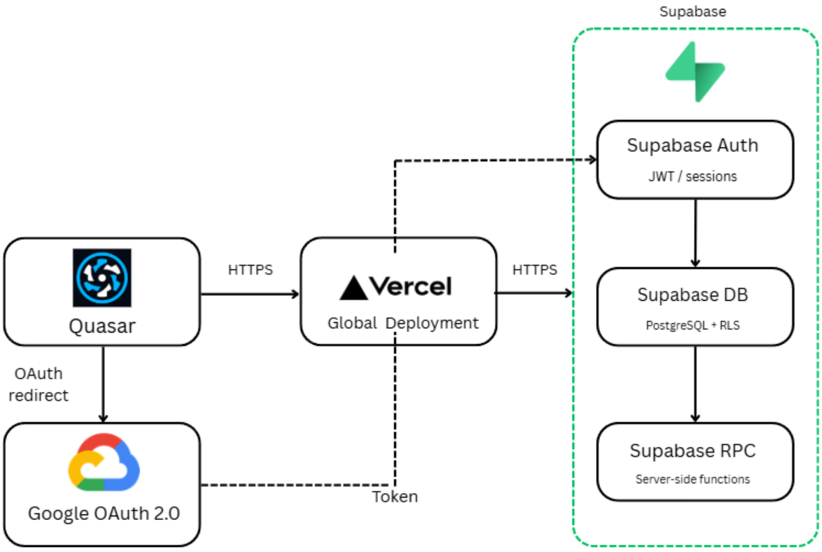

Figure 4.1: System Architecture diagram for CraveControl

4.4 System Authentication

The authentication system CraveControl implements is Supabase Auth [29], which is automatically built into every Supabase project. Supabase Auth supports a multitude of authentication options, with the Google OAuth and the email address and password being implemented for this application. Since Supabase Auth is built into Supabase, it provides a low-friction start to new users and preserves session security in the application. Users are asked to input their age group to contribute to a poll as discussed in Section 4.6.6.

4.4.1 Email and Password Authentication

A method users can register with CraveControl is by using email address and password credentials, with the required field of selecting an age group from a drop down menu. When the account details are submitted, Supabase Auth creates a user record within the authenticated user [29] table which is simultaneously added to the `account` table, alongside the user's selected age group. This permits the user to login to the application using existing credentials registered in the database.

On the login page, there is an option provided to reset an account's password if forgotten. When clicked, the user is prompted to insert the email address linked to the account, which in turn sends an email to the corresponding address using Supabase Auth [29]. This email contains a reset link for the registered account that redirects them to the reset password page, where it validates the reset token sent by Supabase. The new password must be at least eight characters in length and has to be matched in the confirm password input field to be accepted. The user is informed whether the link has expired or is invalid and, if so, will subsequently be redirected to the landing page.

4.4.2 Google OAuth

Google OAuth [28], which is integrated in Google Cloud Services, is another method users can use to create an account with the application. Users that select the Google sign in option will be redirected to a Google authentication screen to complete registration. When confirmed, the user is returned to the application by an authentication callback route. This page extracts the access tokens attached in the URL hash and starts a Supabase session connection. This checks if the email address already exists within the `account` table.

If the email address has not previously been registered, the user is redirected to the age group selection page to ensure that they select an age group before their account creation is finalised. After account creation is completed, the user is redirected to the dashboard page (see Figure 4.2). If the account is already registered, the user is automatically redirected to the dashboard page immediately.

4.5 Database Design

The database attached to CraveControl is hosted on Supabase and contains three relational tables:

- `account`
- `craving_level`
- `daily_purchases`

Each of the three tables is linked to a Supabase authenticated user which is accessed through the `user_id` foreign key labelled as `FK` in the database tables shown below. Row-level security policies enforce data isolation between user accounts. Each table's schemas are provided below in their respective sections.

4.5.1 account

The `account` table stores user account information, including nicotine product details, financial savings total and age group. A new row is inserted when a user registers a new account with their selected age group.

Table 4.2 describes the columns and their role within the `account` table.

Column	Type	Description
`id`	UUID (PK)	Unique identifier for the account record
`created_at`	Timestamptz	Date and time the account was created
`user_id`	UUID (FK)	References the authenticated Supabase user
`age_group`	Text	The user's selected age group, used for contextual information and community statistics
`product_type`	Text	The category of nicotine product used (Disposables, E-Liquids, or Nicotine Pouches)
`product_brand`	Text	The brand of the selected product
`product_name`	Text	The specific product name of selected product
`product_price`	Text	The retail price of the product, used to calculate savings
`product_img_url`	Text	URL of the product image displayed in the application
`total_savings`	Numeric	Accumulated financial savings calculated from daily purchase decisions
`disposal_habit`	Text	The user's self-reported disposal habit, recorded by the anonymous poll on information page

Table 4.2: `account` table schema

4.5.2 craving_level

The `craving_level` table stores the user's individually submitted craving intensity level. A row is created for each logged craving level tracked by the user, storing the relative date.

Table 4.3 describes the columns and their role within the `craving_level` table.

Column	**Type**	**Description**
`id`	UUID (PK)	Unique identifier for the craving entry
`date`	Date	The date on which the craving was logged
`level`	Int2	Craving intensity on a scale of 1 to 5, as selected by the user via an emoji icon interface
`user_id`	UUID (FK)	References the authenticated Supabase user

Table 4.3: `craving_level` table schema

4.5.3 daily_purchases

The `daily_purchases` table tracks the user's overall financial savings accumulated. A question is prompted to the user, whether they purchased their nicotine product that given day. For each time the user answers, a new row is created storing the date and if the user purchased their nicotine product as a boolean value. The boolean value of the purchase determines if the `total_savings` value is updated.

Table 4.4 describes the columns and their role within the `daily_purchases` table.

Column	**Type**	**Description**
`id`	UUID (PK)	Unique identifier for the daily purchase record
`user_id`	UUID (FK)	References the authenticated Supabase user
`date`	Date	The date of the check-in
`purchased`	Bool	True if the user purchased their product that day; false if they resisted
`created_at`	Timestamptz	Date and time the record was inserted

Table 4.4: `daily_purchases` table schema

4.6 Main Features

Below are the core features that CraveControl consists of, each providing its individual unique functionality to the web application.

4.6.1 Dashboard

The dashboard page acts as a home screen for CraveControl. It displays four cards that enable navigation for the entirety of the web application. There is a card for user product selection, which is used for the financial tracking feature, craving intensity tracker, accumulative savings tracker and further information including nicotine helplines and organisations. In the top left corner is a sidebar menu that provides quick access to the user's savings and the product selected (see Figure 4.10). This negates having to navigate to multiple different pages to view personal statistics.

The dashboard is designed to be the main screen the user views on every visit to the application with clear navigation to improve user experience and create a sense of familiarity.

Figure 4.2 shows the layout of the dashboard page alongside its theming that the rest of the application follows to keep consistency.

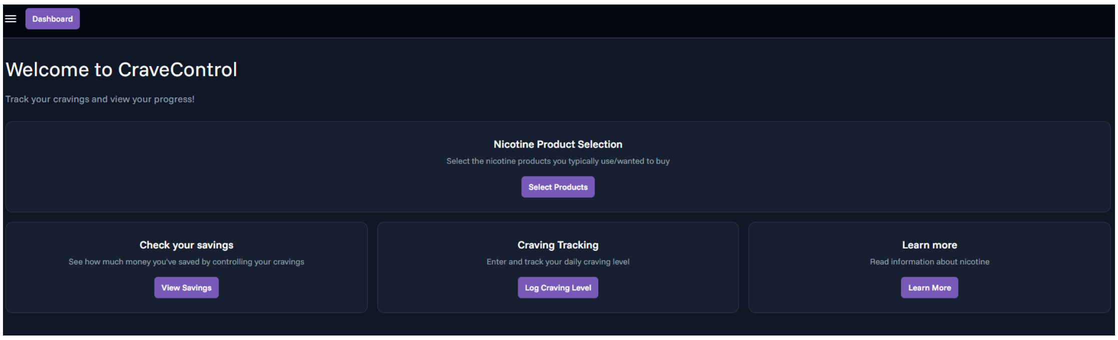

Figure 4.2: Dashboard/Home page for CraveControl.

4.6.2 Product Data Web Scraping

To gather the relevant information pertaining to nicotine products, including product name, image and price, Node.js web scrapers were developed to run on multiple nicotine product websites. Two were developed as part of the application, one for disposable vapes and e-liquids and another for nicotine pouches. *Ecirette.ie* [31] is an Irish vape retailer used for collecting data for vapes and e-liquids. *Snuspal.com* [32] is an international retailer was used for data on nicotine pouches.

The scrapers produced a JSON file organising by the brand name, containing the name, image URL and retail price for each product entry. These are ran once on the developer end, not at deployment or runtime. The JSON files are bundled alongside the application at startup and loaded at runtime. These files are used to populate the product selection page to allow users to select real existing products with realistic pricing to accurately reflect the accumulative savings in the financial tracker.

4.6.3 User Configuration and Setup

Once the user has completed the registration steps and first views the dashboard, the largest card contains the navigation to Product Selection (see Figure 4.2), indicating its importance as the user will be unable to track savings until this field is fulfilled. If the user clicks on the savings page before selecting a product, the savings page will prompt the user to visit the selection page.

In the product selection page the user must complete three steps: select a product type (Disposables, E-Liquids, Nicotine Pouches), product brand and lastly the individual product, as shown in Figure 4.3. Every product in each product type is populated with values gathered from a locally stored JSON file containing the product name, price and image. This data is gathered through a web scraper (see Section 4.6.2). All products have a `product_price` value which stores the product's price which is used for the calculation of overall savings.

Users must click the 'Click to Save' button to bind their desired product to the account.

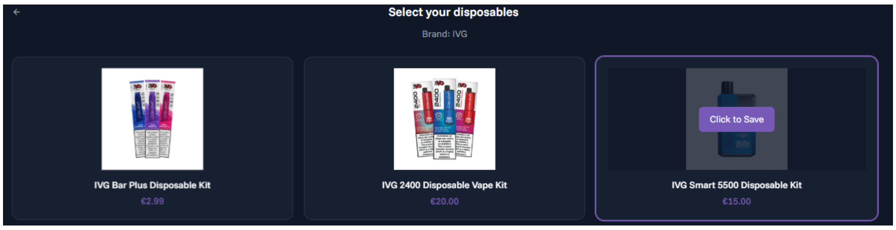

Figure 4.3: A product selection screen within disposables section.

4.6.4 Craving Tracker

The craving tracker enables user to log their daily respective craving intensity using a scale of one to five indicated by different emoji images. Level 1 intensity is depicted as an emoji with a large smile while level 5 intensity is a contrast, being depicted as a frowning emoji, with 'None' on the left side and 'Strong' on the right, as shown in Figure 4.5. The levels in between provide the user with a spectrum of intensity to choose from. Each time the user submits an answer to the prompt, their answer is stored in `craving_level` under the `level` field. When submitted, an upsert operation is utilised with a composite conflict key applied to `user_id` and `date` to ensure only one entry per day is stored, as shown in Figure 4.4 below:

```
await supabase.from('craving_level').upsert(
  {
    user_id: user.id,
    date: new Date().toISOString().split('T')[0],
    level: cravingLevel.value,
  },
  { onConflict: 'user_id,date' }
);
```

Figure 4.4: Upsert operation with composite conflict key

The user is given the opportunity to edit their answer within the calendar day of the original submission date. Instead of duplicate results being stored in the `craving_level` table, the existing result is updated.

Users are unable to view their craving intensity visualisation until they submit an answer to the prompt.

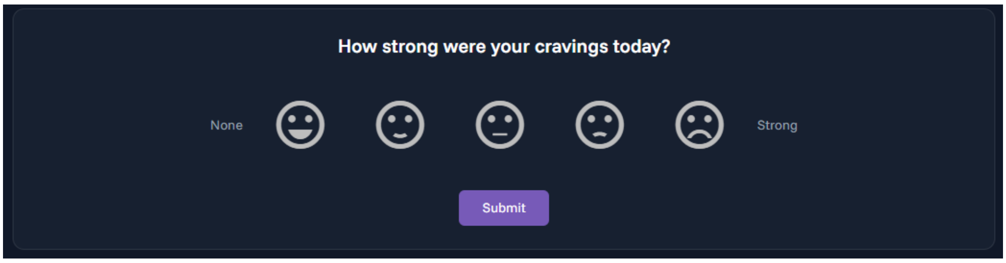

Figure 4.5: Craving tracker using emojis to identify craving intensity.

ApexCharts [13] is used to graphically visualise craving level history data in the form of an interactive line chart implemented in the Quasar frontend. The chart provides a drop down that has the capability to view craving history over three different time period brackets: Current week, previous week and previous month. This allows the user to perform a deeper analysis into their personal craving level patterns across their desired time period.

The x-axis represents the date while the y-axis represents the craving level intensity with the same range as the prompt. If the user does not submit a craving level, it will be represented as a gap in the line chart due to them being null values. This is to prevent data from being misconstrued or misrepresented to the user.

Figure 4.6 displays the line chart with interactive labels that appear when hovering over different points of the chart, displaying different level values for individual days of the week.

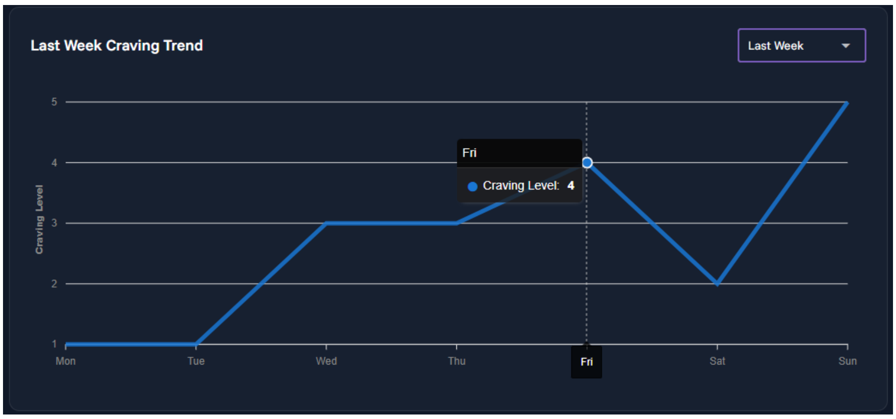

Figure 4.6: ApexChart displaying user's previous week's craving levels.

4.6.5 Financial Savings Tracker

The financial tracker is one of the main components that CraveControl is built upon, designed to motivate users by visualising monetary gains for quitting vaping and nicotine products. Each calendar day, a prompt is displayed asking the user whether they bought the nicotine product attached to their account (see Figure 4.7). If the user has not bought the product and clicks 'No', the `total_savings` balance is incremented by the value stored in `product_price` relevant to their selected product. If the user bought a nicotine product and clicks 'Yes', `total_savings` remains the same and no changes are made. Each result of the purchase prompt is stored in `daily_purchases` table using an upsert operation applied on `user_id` and `date` so only one response is recorded per day to maintain accurate recordings.

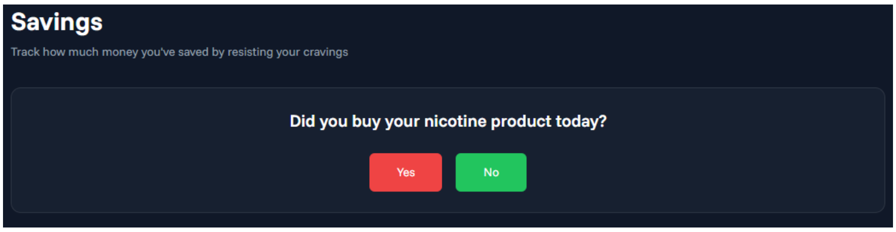

Figure 4.7: Yes/No prompt about nicotine product purchase.

The user is given the option to edit their response on the same calendar day, which is displayed in Figure 4.9. If the user had initially answered 'No' and edits it to 'Yes', the product price is deducted from the total savings. If the user edits their answer to 'No', the total savings amount will increase by the product price. The lowest possible value for total savings has a minimum of zero to prevent the appearance of debt by preventing a negative total, as shown in Figure 4.8:

```
const delta = newAnswer ? -productPrice.value : productPrice.value;
const newSavings = Math.max(0, totalSavings.value + delta);
```

Figure 4.8: Delta-based savings adjustment with zero minimum

The user's accumulative savings total is displayed as text over an image of a savings pot in Figure 4.9. There is an animated effect applied to the text, which starts from zero and increments upwards to the current total value stored in the user account. This provides a visually appealing and engaging method of presenting their personal financial gain.

Figure 4.9: Animated savings pot displaying savings accumulated.

This approach of numeric visualisation provides a concise and accurate calculation of overall personal financial savings tied to the price of the personalised product selected by the user. This method of visualisation overcomes the limitations identified in *Sense2Quit* [1] as *Sense2Quit* provides savings data based on individual cigarette price that requires manual input rather than automatic, personalised incrementation.

4.6.6 Information Page

The information page provides a curated set of resources to aid users in their nicotine cessation. Helplines and organisations linked and mentioned are located in Ireland and internationally, including phone numbers to HSE and NHS services. A link is included to the World Health Organisation [33] quitting resources that provides a collection of international quitting telephone lines and organisations. Each type of nicotine product has their separate section with dedicated links that cover general information, cravings and withdrawal symptoms.

At the top of this page, there is an anonymous poll that users can partake in about the frequency with which personal nicotine products are responsibly disposed of. Each response is stored in the `disposal_habit` column of the `account` table, with the option between 'Always', 'Sometimes' and 'Rarely'. Once the user has submitted a response, the results from other application users are broken down and separated into their respective age groups. Each age group is displayed as a card, with each of the possible answers showing the volume of corresponding responses. These responses are fetched by a Supabase remote procedural call (RPC) function which counts the amount of correlating answers server-side.

This feature provides a community aspect to the application, where users can see how their disposing habits compare to those of both their own age group and others. It adds a factor of environmental awareness to the user about the frequency of appropriately discarding nicotine products.

4.6.7 Sidebar Menu

Throughout each page of the application, in the top left corner, there is a hamburger side menu which provides easy access to the user's email address (username), total savings and the product bound to the account (see Figure 4.10). This feature was implemented as a result of peer feedback in early testing of the application to facilitate a one-click option to view account data without having to navigate to each corresponding page. There is a logout option included at the bottom of the menu, allowing the user to terminate their current authentication session.

These values are retrieved in real time from the `account` table, which is instantly updated after the user submits their answer for the daily purchase prompt or if the user changes their selected product. This is achieved by using custom browser events which are dispatched from the savings and product selection pages for their relative features. The sidebar component is notified of changes from these events, removing the requirement of a complete page reload.

The addition of this menu and ease of access to viewing the savings total amount reinforces the financial motivation aspect of CraveControl without requiring further user interaction.

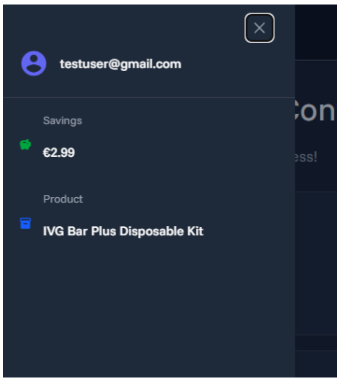

Figure 4.10: Sidebar menu implemented from user feedback.

4.7 UI and UX

For the development of the user interface of CraveControl, Quasar Framework's built in component library [24] was applied which follows the Material Design specification [34]. Quasar's pre-built components which include cards, buttons and input fields, maintained a consistent visual template across the application without having to create custom components manually. Tailwind CSS [35] was implemented to all visible components to create a consistent, custom style across the entirety of the application. It delivers a utility-first concept of styling which allows adjustments and styling to be applied directly to components, eliminating the need for additional CSS files.

The application renders a light and dark mode dynamically, applying the style that matches the device default setting by checking it against the Quasar dark mode state. This ensures that the application matches the theme chosen by the user for the entire device. The accent colour stays a consistent purple throughout the application despite the theme applied which provides continuity regardless of which device CraveControl is opened on.

The principle behind the design of the application is to provide a clean, simplistic appearance which was prioritised after the review of the edits made to the data visualisation in the graph displayed in *Sense2Quit* [1, 21, 16] to reduce visual noise. The emoji icons used to scale the user's level of craving intensity depicted the level value without having to be displayed in numerical digits. The simplicity of clicking either 'Yes' or 'No' for the daily savings prompt improves usability and lowers the barrier to daily engagement.

ApexCharts [13] was chosen for the visualisation of craving intensity data as it has native Vue.js component integration which responds as intended with the Quasar frontend and its mobile compatible rendering. The label indicating the craving level when hovered over dynamically changes to match the theming of the application, providing consistency.

4.8 Development Challenges Encountered

During the development of CraveControl, there was a multitude of challenges faced when experimenting with different functionalities and implementing multiple technologies. Below are the main challenges and obstacles that appeared, especially at the beginning of development.

4.8.1 Local Development Configuration

A major challenge encountered during the early development of CraveControl was configuring the local development environment. To create a local connection to the Supabase CLI, Docker [36] was required to construct isolated service containers necessary for the PostgreSQL tables. However, during the initial setup, Docker failed to create any of the necessary containers. After searching for different solutions in documentation and online discussions, the cause for this failure was identified to be due to virtualisation being disabled in the local device's BIOS.

Docker Desktop for Windows [36], in this case, requires hardware virtualisation to be enabled in the system to allow Linux-based containers to run within the Windows Subsystem for Linux 2 (WSL2) backend. This was not made very clear in the documentation. To solve this issue, virtualisation was enabled in the system BIOS settings which, for the local device, was named AMD-V [37].

4.8.2 Row-Level Security Configuration

Another significant challenge was correctly applying and configuring row-level security (RLS) policies in Supabase [25]. RLS is integrated to all Supabase tables by default [27]. This prevents data from being written or read unless specific policies are defined. At the beginning of development, multiple queries to the database returned empty results, even when data was present in the tables. The issue was identified to be incorrect or missing RLS policies that prevented correct queries from matching any rows. This issue was fixed after policies were defined to the database tables that restricted access to rows where the `user_id` matched with the authenticated user's identifier. This was provided by Supabase Auth [29] as part of the JWT payload (see Section 4.3). This prevented users from having the capabilities to edit user data other than their own.

4.9 Summary

This chapter has demonstrated the design and implementation of CraveControl. It describes the technical decisions made for the system architecture, authentication, database layout and the core features. The challenges encountered during development, including Docker Desktop virtualisation and Supabase row-level security, show the difficulties faced when building a full-stack web application that utilises cloud deployment. The application is built using the Quasar Framework and TailwindCSS for the frontend development, with Supabase for the backend. The application was deployed on Vercel, which rebuilds the application after each new push to the GitHub [7] repository. User authentication was verified using Google OAuth and by using email address and password credentials. There are three databases that support the core features of the application: `account`, `craving_level` and `daily_purchases`. The core components of CraveControl are the craving level tracking, financial tracking, information page and product selection.

Chapter 5

System Evaluation

The chapter compares CraveControl against the declared objectives which determines the completeness of the functionality of the application. This includes the usability for user experience and the effectiveness of the core features. To evaluate these features, two types of tests were conducted: functional testing performed by the developer who knew the expected behaviour and peer-based user testing with a lack of inside knowledge of the implementation of features. The user feedback helped with further improvement and additional features being added into the implementation of the application. The following sections discuss the methodology, testing results and limitations of CraveControl.

5.1 Evaluation Methodology

5.1.1 Developer Functional Testing

The first and main method of evaluation conducted was the developer testing each feature of CraveControl incrementally against its expected behaviour. Due to the developer having access to the code implementation and knowledge of the intended output, this allowed for a systematic verification of each component's correct functionality under typical conditions. Every core feature was extensively tested against its intended output, including account authentication, savings calculation and chart visualisation.

This approach was significantly useful for the verification of correct functionality, as the tester (developer) can create test inputs to validate edge cases and confirm the application's system responds as expected. However, since the developer has familiarity with the inner workings of the system, this creates a bias that may cause unconscious avoidance of unexpected interactions that would cause usability issues to arise. This limitation was overcome by implementing peer-based user testing which showcased the use of the application from a first-time user experience, which is discussed below in Section 5.1.2.

5.1.2 Peer-Based User Testing

The second method of testing was based on a group of four participants (n=4) who had no prior knowledge of the inner implementation of CraveControl interacting with the application, then giving feedback on their user experience. Participants ranged in age from 22 to 55 and included both nicotine users and non-users which helped to provide a range of perspectives. The group was asked to navigate throughout the application by themselves with no given instructions or directions to keep the testing as authentic as possible. Users provided feedback through informal discussions following each test run which covered navigation clarity, feature interactions and visual appearance of the application.

There were two stages where peer testing was conducted. The first round was performed during the development process, but still incomplete, before and after it was deployed onto the Vercel cloud platform [30] for ease of access. After feedback was received from this stage, there were numerous design improvements that were implemented before the final version of the application was released. The second round of testing was conducted on the completed final version of the application to see the effectiveness of the improvements. This iterative approach was used as it focused on user-centric design principles, where feedback was incrementally and progressively added throughout the development of CraveControl.

While the sample size was small and feedback was collected informally rather than through a structured formal user study, it provided valid insight into how users would typically be expected to interact with the application's design and interface with no prior knowledge. This approach is consistent with the evaluation approach conducted by similar cessation applications like *Sense2Quit*, where user feedback was utilised to make changes to the application design.

5.2 Functional Testing Results

After the two types of testing were performed, CraveControl's core features were discovered to be fully functional. Table 5.1 provides a summary of the features tested and their outcome.

All features passed the functional testing which demonstrates the complete functionality of the application and that it behaves as intended. There were no critical bugs or data issues identified in this phase of development.

Feature	**Expected behaviour**	**Outcome**
Google OAuth login	User authenticated via Google account and session created	Pass
Account setup flow	User selects product type, brand, name and age group; data saved to `account` table	Pass
Product price stored	Selected product price persisted to `account.product_price` for use in savings calculation	Pass
Craving level submission	Craving level and date saved to `craving_level` table under authenticated `user_id`	Pass
Craving chart rendering	ApexCharts line chart populated with user's craving history, ordered by date	Pass
Daily purchase check-in	Yes/No response saved to `daily_purchases` table; savings updated if 'No' selected	Pass
Savings accumulation	`total_savings` in `account` table incremented by `product_price` on each 'No' response	Pass
Dashboard display	Cards with correct navigation to rest of application	Pass
Sidebar quick stats	Sidebar displays current savings, selected product name, and username correctly, logout displayed at bottom	Pass
Support resources page	Helpline information and links displayed correctly with respective titles	Pass
Row-level security policies	User can only access and modify their own records across all tables	Pass

Table 5.1: Functional testing results

5.3 Peer Testing Results

5.3.1 General Feedback

After the peer testing was conducted, the feedback was predominatly positive across both rounds of testing. Participants responded well to the visual design and appearance of the application, noting the clean and concise user interface for navigation. Using Quasar Framework's [24] component library provided a consistent visual layout that added familiarity. The dashboard in particular was noted to be an effective home page, providing clear access to all core features of the application.

5.3.2 Iterative Improvement: Sidebar Quick Statistics

In the initial round of testing, one user suggested a quick persistent summary option be provided to the user to remove the need for navigation to multiple pages just to view their current progress. Specifically, the user suggested a sidebar menu to implement this which would display the user's current savings, username and selected product. The idea was that without having to navigate to the dashboard and the relevant pages, users could view their progress from wherever in the application.

This was directly acted on, and the quick stats sidebar menu was implemented with the final version of the application. The panel displays the user's username, total savings and selected product. These values were retrieved in real-time from the `account` table in the Supabase database. The 'Logout' functionality was added to this sidebar menu at the bottom to remove clutter from the header of the dashboard. This addition was validated in future peer testing, where the change was accepted positively. The persistent availability of the savings total throughout the application reinforces the financial motivation aspect of CraveControl without requiring additional navigation.

This additional functionality discovered through user testing highlights the value of early-stage peer testing in identifying missing or underspecified features that the developer had not conceptualised by having familiarity with the system. It demonstrates the value of the financial savings feature, when it is constantly available throughout the application instead of being restricted to a single page. This is a principle supported by papers on positive reinforcement in health cessation applications [1, 2, 21].

5.3.3 Usability Issue: Product Save Button

One usability issue that arose during peer testing was the 'Save' button when attempting to save a product to the user's account. When the user selected a product to save to their account to begin their savings calculation, a 'Save' button appeared over the desired product card to confirm selection. Despite the button indicating it needed to be clicked to save the product as it was in present tense, the user assumed that the initial click automatically bound this product to their account and required no further action. As a result, when the user tried to start their savings journey, the savings page told the user to visit the product selection page before they could begin, meaning the product did not persist.

This issue relates to Nielsen's usability principle of visibility of system status [8], which states that a system should always keep users informed about what is currently happening through relevant feedback from the system within an appropriate time frame. For this case, the page interface did not appropriately communicate to the user that further action was required to confirm their product selection. The assumption created by the action of selecting a product that would automatically be updated once clicked is reasonable due to the prevalence of auto-save behaviours found in modern applications.

A recommended solution to this issue in a future iteration of the application was to implement a more specific visual indicator that clearly indicates to the user that there is further action needed to save the product to their account. This finding demonstrates the value of peer-based testing in surfacing usability assumptions that developer testing alone would not reveal.

5.4 Strengths

Below discusses the overall strengths of the application, including the usability and rundown of each feature implemented into the application.

5.4.1 Overall Nicotine Focus

CraveControl is designed with overall nicotine cessation in mind which distinguishes it from the majority of existing cessation applications, which mainly target smoking. As discussed in Chapter 3, none of the similar cessation applications reviewed were developed in aid of overall nicotine product cessation and behaviour. By targeting the application towards nicotine products including characteristics like product types, pricing and tracking consumption patterns, CraveControl provides more relevant and accurate saving calculations and craving tracking.

5.4.2 Personalised Financial Tracking

The financial tracker functionality of CraveControl is different to the other cessation applications reviewed as it uses the user's specific product and its corresponding price to calculate the accumulative savings. Instead of using a generic average cost, the application calls on the `product_price` value stored in `account` table to calculate an accurate, personalised representation of financial benefit of cessation. This uses real pricing for the user's nicotine product to keep as realistic as possible. This makes the feature more effective as a motivation tool than if a generic value was used. This addresses a limitation encountered in *Sense2Quit* [1, 21] where the data for the financial savings was based on manual input of the cost of each individual cigarette.

5.4.3 Accessibility and Ease of Use

CraveControl requires no additional account migration, hardware or technical aspects besides the Google account sign in function to use the application. Building the application using a framework like Quasar Framework [24] ensures that the tool is accessible from both mobile and desktop devices without an installation process. The flow of using the application is concise, where the daily interactions are minimal, in both use and navigation, to reduce the effort needed to engage with tools in the cessation attempt.

Unlike *Sense2Quit* [1, 21] and *QuitGenius* [2] which are both native mobile applications that require installation to the device, CraveControl is built as a web application that avoids an installation process. This helps to increase its accessibility to a wider user base.

5.4.4 Interactive Data Visualisation

ApexCharts [13] was used to render the craving intensity history as an interactive line chart that allows users to engage with their personally logged data. The line chart supports hover-based functionality that displays the date and the corresponding craving level for each logged entry, unlike the static bar chart used in *Sense2Quit* [1, 21] and the statement based display in *QuitGenius* [2]. The information displayed when hovering over the line chart prevents the rest of the page from being cluttered while providing accurate data. The time-based chart makes the craving intensity trends more visually apparent over time which provides users with self-awareness of daily behaviour patterns for approaching cessation. This is consistent with survey findings that identified line charts as most preferred for health visualisation [16].

5.4.5 Iterative Development Responsiveness

By listening and implementing features suggested by peer feedback before the final version of the application shows a responsiveness to user needs that aided the quality of CraveControl to the typical user. The sidebar menu showing quick statistics that was implemented as a result of peer testing is a strong example of user-centric design in the application which aligns with principles of iterative software development.

5.5 Limitations

Below are the limitations encountered during the development of CraveControl and at the end of the development period.

5.5.1 Single Chart Type

The visualisation of craving data in CraveControl is currently only available as a line chart. While this chart type is well suited for displaying craving intensity over time, additional chart types could be utilised to provide more perspectives on the user's data. An example is a bar chart using a calculated average of craving levels for each day of the week that could also aid in identifying patterns in craving behaviour. In future iterations of the application, the availability of multiple chart types would increase the analytical value of the visualisation section [16] and would make it similar to the multi-format approach depicted by *QuitGenius* [2].

5.5.2 Manual Daily Input

The application completely requires daily manual input from the user, including the financial monitoring and craving tracker. This shows the dependency the application has on consistent user engagement which may be difficult to uphold over the duration of a longer quit attempt. In the case that the user forgets to log a craving or answer the savings prompt, the data is lost and cannot be added on a future date. This can lead to gaps in the visualisation and underestimate the actual savings. The absence of push notification or reminders means the application does not prompt the user to engage daily, which is a limitation compared to native mobile applications that have access to the device's notification system.

5.5.3 No Wearable/Sensor Integration

Wearable health monitoring systems have the capabilities of capturing physiological data that corresponds with nicotine use, including changes in heart rate and oxygen levels, as discussed in Section 3.3. CraveControl does not involve any sensor-based data, showing it solely relies on self-reported data. This decision was made for wider accessibility, however this results in the application lacking in providing objective physiological feedback personalised to the user. In a future version of the application, the integration of wearable sensor data could provide additional data that complements the self-tracked craving levels.

5.6 Summary

This chapter evaluated CraveControl through developer functional testing and two rounds of peer-based user testing. All main features of the application were confirmed to be fully functional after testing. User feedback was overall positive, especially regarding the application's visual design and navigation layout. The key findings from peer testing was the suggestion and eventual implementation of a sidebar menu displaying quick statistics and a usability issue with the product save button. The general nicotine focus, the personalised financial monitoring and interactive visualisation were the strengths of the application. Its main limitations are providing one type of visualisation, dependency of manual daily input and lack of wearable sensor integration.

Chapter 6

Conclusion

6.1 Overview

This paper investigated CraveControl, a web application developed to provide support in nicotine cessation through two complementary systems: the visualisation of self-tracked craving levels and the monitoring of accumulative financial savings through avoided purchases. The reasoning behind this research was due to the rising prevalence of vaping and nicotine pouch consumption in younger generations, alongside a personal awareness of the difficulties those people face in the attempts to quit. Through this research, the lack of cessation support tools that are not traditional smoking orientated was uncovered, which contributes to the challenge of quitting without technical support tools. This chapter reflects on the findings of the paper across each chapter, including obstacles encountered during the research and development process, and describes possible directions for future work.

6.2 Summary of Findings

The basis of this paper was to investigate different approaches to tracking personal craving patterns and financial spending of people who regularly consume and purchase nicotine products, while displaying the data to the user in a readable, clear manner. The methods utilised to achieve this were found to be of multiple types: graphical, numerical, qualitative or inclusive of all three, demonstrated by similar applications in Chapter 3.

In response to RQ1 (see Chapter 1), ApexCharts [13] was identified as the most appropriate visualisation library for CraveControl as it provides interactive SVG based charts with Vue.js integration and allows for hover-based chart inspection. This is backed by research stating that mobile health users preferred simple visualisation over complex methods [16]. Peer testing validated that the chart was easy to use and navigate without instructions.

In response to RQ2, the line chart was discovered to be the most suitable technique for displaying self-tracked craving levels, with dates on the x-axis and craving levels on the y-axis. Null values were displayed as gaps to prevent misinterpretation. This is consistent with findings that indicate line charts are one of the most desired chart types in mobile health applications [16, 9].

In response to RQ3, a product-based calculation tied to the user's selected product price was found to be the most effective approach to financial monitoring. This provided personalised saving accuracy instead of generic estimates used by *QuitGenius* [2] and the manual input method used by *Sense2Quit* [1]. The animated savings pot and persistent sidebar menu provide an engaging and easily accessible representation of accumulated financial savings.

A significant finding from the technology review was the absence of cessation applications dedicated to nicotine products, excluding traditional smoking. The search found no applications targeted or designed for vaping or nicotine pouch cessation, despite their popularity and normality among younger demographics. The reviewed applications, *Sense2Quit* and *QuitGenius*, exclusively target traditional smoking as discussed in Chapter 3, and represent a field that has not expanded alongside the increase of the consumption of these nicotine products besides cigarettes.

Outside of the gap in general nicotine cessation tools, there were further gaps identified within existing applications. None of the reviewed applications provide both craving intensity tracking with personalised, product based accumulative financial tracking. *Sense2Quit* provided financial calculations based on manual input of the price of a box and finding the individual cost of a single cigarette, instead of a specified product. *QuitGenius* provided a generic financial estimate rather than a personalised product based calculation. It operates on a business-to-business model that restricts access to users within an organisation that own a license. Neither of these applications tracked craving levels to provide pattern feedback over time. *HOT Watch* operated as a passive wearable monitor that collected physiological data relevant to nicotine consumption, however it did not include any cessation-based functionality and introduced a hardware dependency. This weakened its accessibility to a wider audience.

There was a highlighted lack of technical research papers for personal financial monitoring systems within and outside a cessation context. The majority of papers discovered covered large-scale industrial datasets and fraud detection instead of personal spending tracking relating to behaviour change. There were no technical papers found to explore the correlation between personal financial monitoring and data visualisation, indicating an underexplored area of research that CraveControl begins to address.

CraveControl is designed and implemented to address these identified gaps. The application is built using the Quasar Framework alongside Tailwind CSS for frontend design, with a Supabase backend and authentication service. It is deployed on Vercel and users can authenticate an account using email address and password credentials or Google OAuth. The application provides a fully functional, accessible web cessation tool that requires no additional hardware devices. Product data was captured using Node.js web scrapers that used ecirette.ie for vape products and snuspal.com for nicotine pouches. These produced structured JSON files that were utilised in populating product selection with real products with their corresponding price. Its three main database tables, `account`, `craving_level`, `daily_purchases`, support overall flow and functionality of the application. The profile details, including age group and total savings, and the selected product details are handled by the `account` table. The craving tracking information, which can be viewed on the interactive ApexCharts line chart with time filters, is handled by the `craving_level` table. The prompt for product purchases to contribute to the accumulative savings is handled with help from the `daily_purchases` table. A sidebar menu accessible from any page that was implemented as a response to user feedback allows the user to view their account details from anywhere within the application. All technologies used are described in Chapter 4.

During the evaluation of the application, it was confirmed that all core features were completely functional. Peer-based feedback was generally positive, complementing the visual design and clear navigation. With suggestions of features like the sidebar menu that were later implemented into the application, this shows a user-based development response, with user experience at the forefront of development. In line with iterative software development principles, a usability issue was identified in the product selection page where lack of clarity on selection feedback caused a user to assume the product had been automatically saved. This was fixed in later development, which indicated the importance of user testing.

6.3 Obstacles and Challenges Encountered

Multiple obstacles were encountered during the research and development phases of CraveControl. In the research phase, the near absence of academic papers regarding personal financial tracking systems meant that this part of CraveControl's functionality gathered information from limited examples like *Sense2Quit*. Another challenge in researching for this project was the disparity of literature on cessation applications that addressed vaping or nicotine pouches. The majority of papers found focused on traditional smoking, which required further assessment into whether or not the findings were transferable to an overall nicotine context, or at least vaping.

During the development phase, a major challenge encountered was the configuration of the local development environment. To run a local instance of Supabase using Docker required hardware virtualisation to be enabled in the device BIOS, which was not initially made clear in the documentation as part of the requirements. It was later identified over multiple secondary sources through independent research. Another challenge involved the correct configuration of the row-level security policies in Supabase, where missing or incorrect policy declarations caused empty results to be returned from queries in early development. The iterative feedback system during development required the application design to be flexible, especially in the addition of the sidebar menu which introduced persistent real-time data fetching requirement across all application pages.

6.4 Future Work

There are multiple directions for research that could extend the capabilities of CraveControl and increase its contribution to the research field. The most immediate improvement would be a modification made to the save button usability issue in the product selection page identified in Chapter 5, by implementing automatic product saving. This would eliminate a point of confusion in the application flow and reduce the risk of the financial tracker not functioning due to an unsaved product selection.

There could be more chart types in aid of the visualisation aspect of the application. One of these charts could be a bar chart displaying the average craving level for each day of the week or a calendar heat map of craving intensity. This would provide users with further perspective into their craving habits and behaviour, which is consistent with research indicating mobile health application users preferring a combination of chart types for health data. The inclusion of push notifications or daily reminders would address the current reliance on voluntary engagement, which may have negative effects on a quit attempt.

The addition of wearable sensor hardware integration could provide additional insight that complements the self-reported craving data collected by CraveControl since devices like *HOT Watch* collects physiological indicators that relate to nicotine consumption and use.

Since there is an evident lack of papers investigating the correlation between personal financial monitoring and data visualisation systems, if there is further research into this area it could increase the evidence base for this approach and produce stronger academic grounding into future iterations of CraveControl. Similarly, as data visualisation techniques are constantly expanding and emerging, the visualisation feature of the application could be expanded to include more advanced approaches as they become more relevant.

6.5 Concluding Remarks

CraveControl represents a meaningful contribution towards the gap in technical support tools for those who wish to quit vaping and nicotine products. It is an easily accessible web application that is distinct from smoking-centric cessation applications that currently dominate this field. It combines self-tracked craving visualisation with personalised accumulative financial monitoring, available on any device connected to the internet. In the research process, it was identified that, not only is there a lack of nicotine cessation applications but an absence of personal financial monitoring systems in health behaviour change. This reinforces the novelty of CraveControl's approach to nicotine cessation. The development process demonstrated an iterative, user-based approach in producing a functional and responsive tool. It is hoped that this work contributes to a growing recognition to vape and nicotine product cessation as a field that needs further dedicated research and technological investment.

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